

Airbnb Exploratory Data Analysis

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1 Introduction

Short-term rental markets are shaped by a combination of geography, property characteristics, host behavior, and local market conditions. Although platforms such as Airbnb provide a common marketplace across cities, the listings within each market may differ substantially in price, spatial distribution, host composition, room type, and availability. Comparing these patterns across multiple cities provides an opportunity to identify both broad similarities and important local differences.

This report presents an exploratory analysis of Airbnb listings across five U.S. markets: Austin, Chicago, Los Angeles, New York, and Portland. Using September 2025 data provided by Inside Airbnb, the analysis examines several questions: How do nightly prices vary across and within cities? How are listings distributed among individual, small, and large hosts? How does room composition differ geographically and across host categories? Which listing characteristics are associated with changes in nightly price? Finally, do patterns in listing availability appear to be more closely associated with host type, room type, or geography?

The analysis combines data cleaning and feature engineering with descriptive statistics, spatial clustering, and regression modeling. DBSCAN is used to identify geographically concentrated groups of listings without imposing predetermined neighborhood boundaries, while ordinary least-squares regression is used to examine associations between listing characteristics and nightly price for entire-home and apartment listings. New York is also examined at the borough level because its market structure is not fully represented by the broader clustering analysis.

The goal of this project is not to construct a predictive pricing or booking model, but to develop an interpretable comparison of five distinct short-term rental markets. Particular attention is given to where broad patterns hold across cities, where individual markets depart from those patterns, and which conclusions are supported by the available data.

2 Data Preparation

All listing data were obtained from the September 2025 Inside Airbnb releases. In accordance with the Inside Airbnb data-use request, the raw CSV files are not distributed with this project and must be downloaded separately from the source. Population estimates used for cross-city context were obtained from the U.S. Census Bureau.

The cleaning process standardized the variables required for analysis, converted price and other numeric fields to appropriate data types, encoded selected categorical variables, and derived several features used throughout the report. The amenities field, for example, was reduced from a heterogeneous list of text descriptions to the number of amenities reported by each listing.

The dataset also contained a small number of observations with internally contradictory or implausible combinations of attributes. Examples included extremely high nightly prices that appeared to function as booking deterrents and inactive listings with little supporting information. A rule-based flagging procedure was used to identify suspicious observations. Listings were removed only when they violated hard-pass rules or accumulated multiple contradictory indicators; high price alone was not sufficient for exclusion. This approach was intended to reduce distortion without automatically removing legitimate high-end listings.

2.1 Spatial Clustering

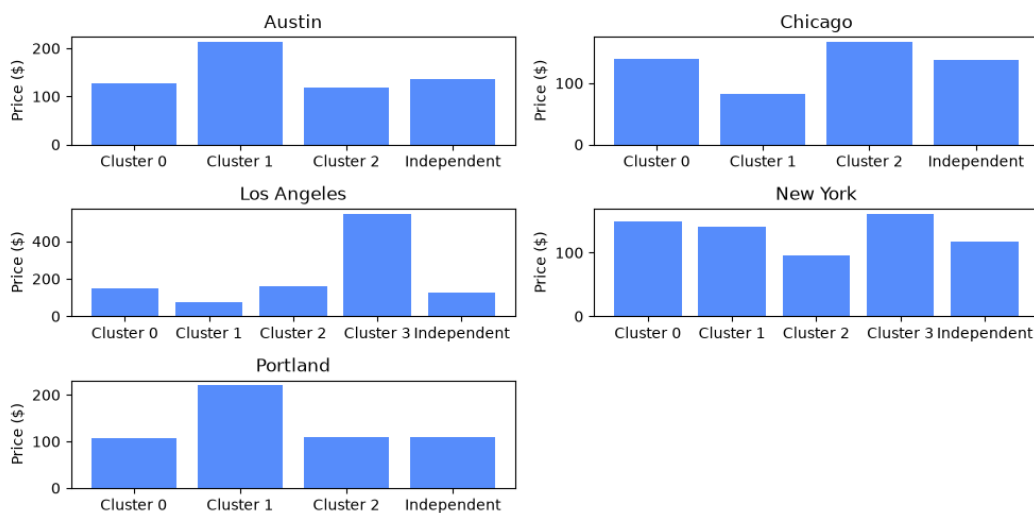
Listings are not distributed uniformly within cities. Population centers, geography, tourism, and housing patterns create spatial concentrations that may operate as distinct local markets. To identify these concentrations without manually imposing neighborhood boundaries, the analysis uses DBSCAN.

DBSCAN was selected because it can identify arbitrarily shaped clusters and label isolated observations as noise. The `eps` and `min_samples` parameters were selected separately for each city through iterative evaluation of cluster stability and geographic interpretability. The objective was to identify meaningful population centers without unnecessarily fragmenting larger spatial clusters. Listings labeled as noise by DBSCAN are referred to as independent listings throughout the analysis.

3 Results

3.1 Pricing

Cluster prices are generally consistent within each city, although most markets contain at least one cluster with a noticeably higher or lower median nightly price. These differences are consistent with localized influences such as tourism, neighborhood affluence, housing stock, or limited supply. The remaining clusters tend to exhibit comparable median prices, suggesting that nearby listings often operate under similar market conditions.

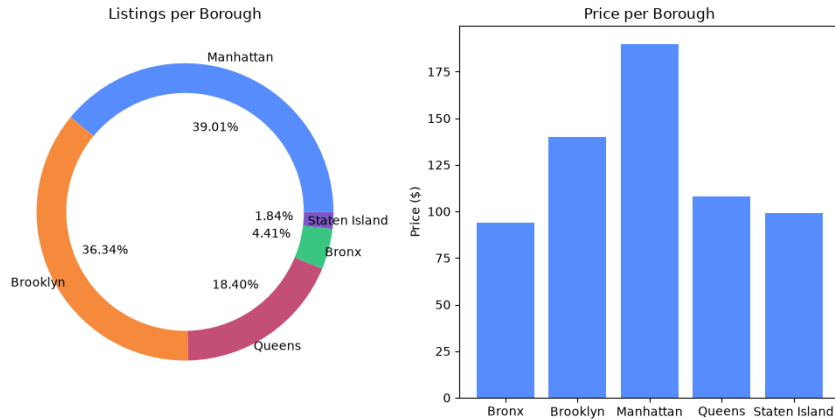


- **Austin:** Cluster 1, located near the Hudson Bend and Lakeway communities, exhibits the highest median nightly price. Proximity to Lake Travis and higher surrounding property values

may contribute to this increase.

- **Chicago:** Cluster 1, surrounding the University of Chicago, exhibits lower prices than the remaining clusters. This may reflect differences in the surrounding housing stock or a local market serving university visitors.
- **Los Angeles:** Cluster 3, which encompasses the majority of Santa Catalina Island, exhibits substantially higher median prices than the remaining clusters. Limited housing supply and strong tourism demand are plausible contributors. Clusters 1 and 2 exhibit relatively small coefficients of variation, while Cluster 0 displays substantially greater variability, suggesting that neighborhood-level analysis would provide additional insight.
- **New York:** The Staten Island clusters exhibit lower median prices than much of the broader New York market, suggesting that population density alone does not explain local price variation and that factors such as tourism demand and proximity to major visitor destinations may play a larger role.
- **Portland:** Cluster 2 encompasses the Hillside and Arlington Heights neighborhoods, which are generally more affluent than those in the remaining clusters. Correspondingly, listings in this cluster exhibit the highest median nightly price.

New York’s borough structure provides a useful level of detail not available in the other city-level comparisons. Cluster 0 contains listings from four of the five boroughs, while Clusters 1 and 2 are concentrated in Staten Island. A borough-specific breakdown therefore provides a more informative view of listing shares and prices.



Manhattan and Brooklyn account for the majority of New York listings, with the remaining share divided among Queens, the Bronx, and Staten Island. Manhattan exhibits the highest median nightly price, followed by Brooklyn. These results are consistent with the distinct housing and tourism markets of the two boroughs, although the available data do not isolate the causal mechanisms behind the price differences.

Across the five markets, median nightly prices differ noticeably, with Portland exhibiting the lowest median and Los Angeles the highest. Price variability also differs across markets. New York and Portland have the lowest coefficients of variation, indicating that prices are more tightly centered relative to their means, while Austin, Chicago, and Los Angeles exhibit greater relative dispersion.

The five-city comparison suggests that larger markets tend to exhibit higher median nightly prices. However, with only five cities, this relationship should be interpreted as descriptive rather than as evidence of a broader statistical association. Portland and Austin are particularly notable because their median prices differ by approximately \$20 despite being more comparable in population size than the largest markets.

3.2 Host Analysis

Across all five cities, hosts range from individuals managing a single listing to organizations operating many listings. Hosts were classified according to the number of active listings associated with each host:

- **Individual Host:** one listing;
- **Small Host:** between 2 and 9 listings;
- **Large Host:** 10 or more listings.

These categories describe the number of listings associated with a host, not necessarily the number of distinct properties. Multiple listings may represent separate properties, multiple rooms within one property, or a combination of both.

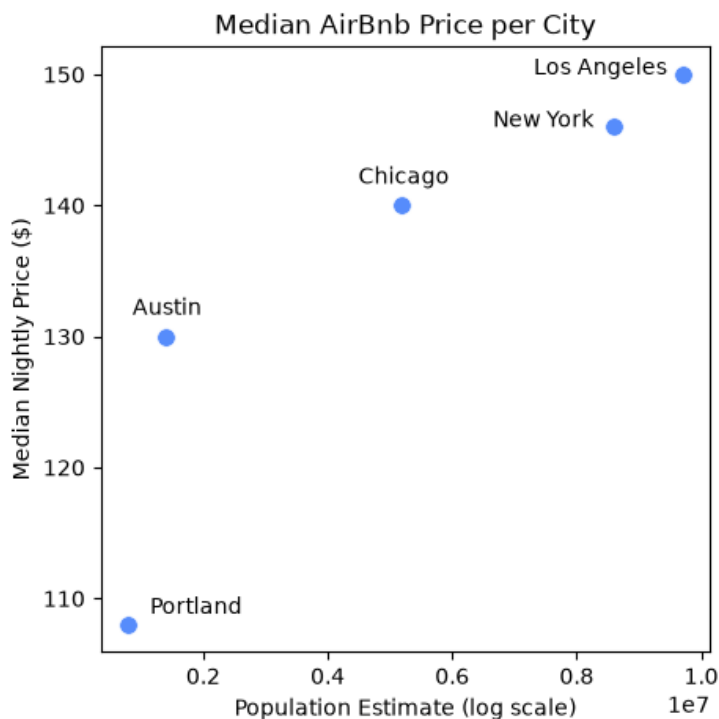


Individual hosts are the most prevalent group in every city, accounting for approximately 70% to 82% of the host population. Small hosts constitute the second-largest category, while large hosts represent only a small fraction of hosts. New York and Portland exhibit especially small shares of multi-listing hosts. Local short-term rental regulations may contribute to these patterns, although regulatory explanations should be interpreted as contextual rather than causal unless supported by additional policy analysis.

Despite representing a much smaller fraction of hosts, small hosts control a share of listings comparable to that of individual hosts in several cities. This indicates that the typical small host manages multiple active listings, while relatively few hosts operate near the upper limit of the category.

Portland departs from the pattern observed in the other cities, with individual hosts accounting for a larger share of listings. One possible explanation is the city’s more restrictive short-term rental environment, which may discourage the operation of multiple listings.

Large hosts represent only a small fraction of the host population but control a substantially larger share of listings. This demonstrates the concentration of listings among a relatively small number of high-volume operators.



Small hosts generally exhibit lower median nightly prices than individual and large hosts. The room-composition analysis provides one possible explanation: small hosts contain a larger share of private-room listings, which are typically less expensive than entire properties. The available data do not provide enough information to determine why individual and large hosts tend to exhibit higher prices.

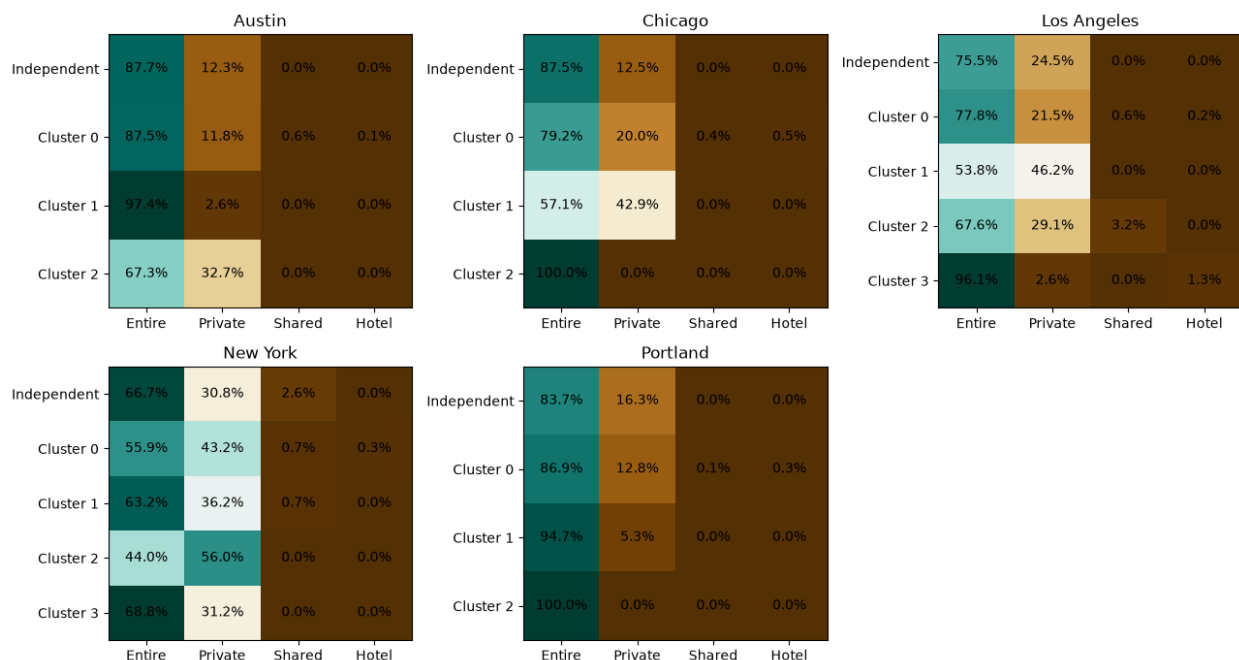
Portland again differs from the other four markets. Despite large hosts representing less than 1% of the host population and approximately 8% of listings, this category exhibits the highest median nightly price among Portland host groups.

At the borough level, host composition is relatively consistent across most of New York, with

Manhattan providing the clearest exception. Large hosts represent a substantially greater share of Manhattan listings than in the other boroughs. At the same time, median nightly prices across Manhattan host categories are relatively similar, suggesting that borough-level market conditions may be more important than host category alone.

3.3 Room Composition

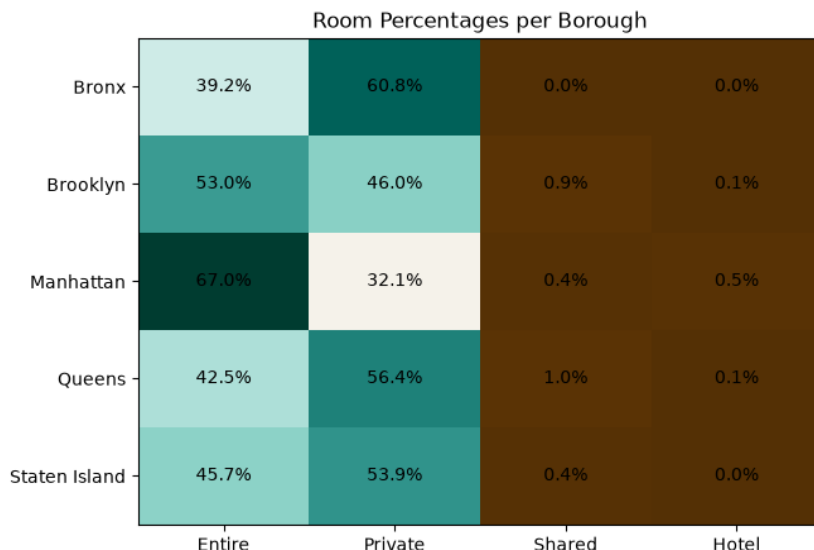
The Inside Airbnb dataset classifies listings into four room types: Entire Home/Apartment, Private Room, Shared Room, and Hotel Room. These categories provide a direct way to compare listing composition across spatial clusters and host groups.



Entire homes and apartments constitute the majority of listings in most markets, while private rooms form the second-largest category. Shared rooms and hotel rooms have little representation in the five-city sample.

- **Austin:** Cluster 2, located near Hornsby Bend, contains a substantially larger share of private rooms than the other Austin clusters.
- **Chicago:** Cluster 1 contains a larger share of private rooms and also exhibits lower median prices than the other clusters. Because private rooms are typically less expensive than entire properties, room composition may partially explain the lower cluster-level median.
- **Los Angeles:** Cluster 1, located around Palmdale, contains a larger share of private rooms than the other Los Angeles clusters.
- **New York:** All clusters contain comparatively large shares of private rooms, and Cluster 2 contains a majority of private-room listings. New York's housing stock and short-term rental environment may contribute to this composition.
- **Portland:** Room composition is relatively consistent across clusters. Independent listings contain a somewhat larger share of private rooms than the major clusters.

Room composition is broadly consistent with the city-level results, but small hosts typically have a larger percentage of private rooms than individual or large hosts. This provides a plausible compositional explanation for the lower median prices observed among small hosts: a greater share of their listings belongs to a generally less expensive room category.



New York's boroughs again reveal meaningful local differences. Manhattan contains a substantially larger share of entire-home listings than the other boroughs. Combined with Manhattan's higher median nightly price and more even distribution of listings across host categories, this helps explain why the borough differs so strongly from the broader New York market.

3.4 Regression Analysis

We next consider which listing characteristics are associated with changes in nightly price. To obtain meaningful coefficient estimates, comparisons should be made among similar listing types. An entire home and a hotel room, for example, serve different markets and are therefore not directly comparable.

Across all five cities, the room-type composition is:

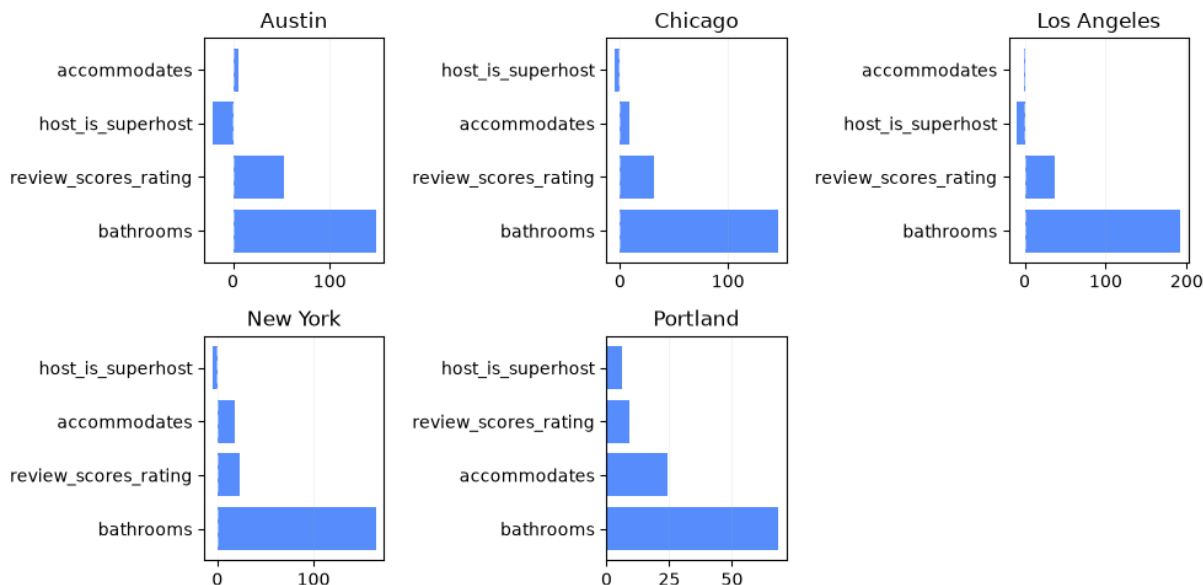
- Entire Home/Apartment: 68.33%;
- Private Room: 30.07%;
- Shared Room: 0.6888%;
- Hotel Room: 0.9017%.

Shared rooms and hotel rooms represent less than 1% of the observations and were excluded from the regression analysis because their small samples would limit the reliability of cross-city comparisons.

Private rooms, while more common, presented a different challenge. Multiple model specifications were explored using different combinations of explanatory variables. Across all five cities, these models consistently exhibited poor explanatory power, few explanatory variables were statistically significant, and the overall models provided limited evidence that the selected characteristics

explained private-room price variation. Consequently, the remaining analysis focuses on Entire Home/Apartment listings, where the regression models produced substantially stronger and more interpretable results.

For Entire Home/Apartment listings, an ordinary least-squares (OLS) regression model was fit independently for each city using the `statsmodels` package. After evaluating multiple model specifications and checking multicollinearity among candidate variables, the final model included superhost status, accommodation capacity, review score, and number of bathrooms.



Across all five cities, the models exhibited stronger explanatory power than the private-room models, with R^2 values ranging from 0.34 to 0.53. Bathrooms and accommodation capacity were statistically significant in every city, whereas review score and superhost status varied across markets.

Bathrooms consistently produced the largest positive coefficients. Although this suggests that additional bathrooms are associated with higher nightly prices, the variable likely also serves as a proxy for overall property size. Because the dataset does not include square footage, bathrooms capture information about property scale that is not otherwise available.

Accommodation capacity was statistically significant in every city but produced comparatively small coefficient estimates. This suggests that the number of guests a property can accommodate contributes less to pricing than broader indicators of property size.

Review score was statistically significant in all cities except Portland. Where significant, a one-unit increase in review score was associated with an estimated \$20 to \$55 increase in nightly price, holding the other model variables constant.

Superhost status was the least consistent variable, reaching statistical significance only in Austin, Los Angeles, and Portland. Its coefficient also varied in direction across markets. This suggests that the association between superhost status and price depends on local market conditions rather than representing a universal pricing relationship.

The number of amenities was also evaluated. Although statistically significant in several cities, its estimated coefficients were negligible, typically less than \$0.25 per additional amenity. The variable was therefore removed from the final model because it added little practical interpretability.

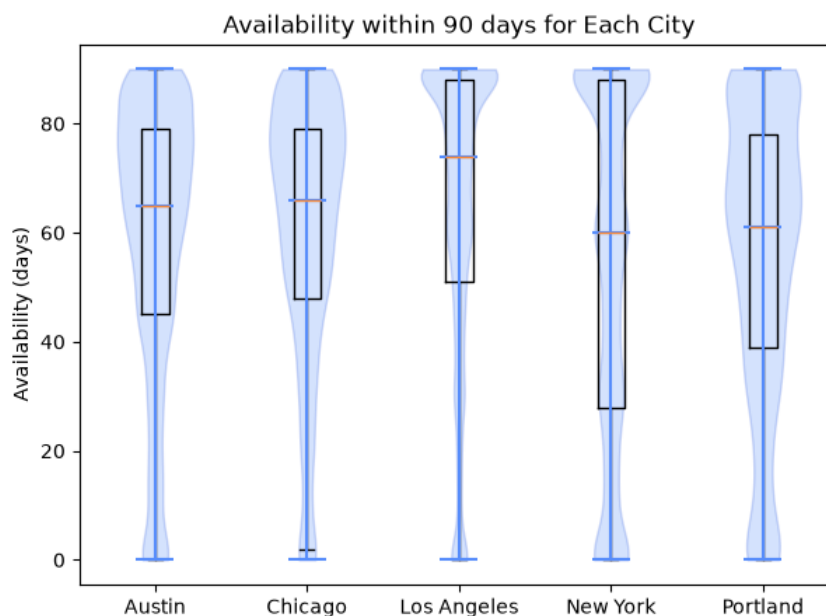
Overall, the regression analysis suggests that characteristics associated with property scale and

perceived quality are more consistently associated with nightly price than host-specific characteristics. The models are explanatory rather than causal: the coefficients describe conditional associations within the available data and should not be interpreted as the effect of changing a listing characteristic in isolation.

3.5 Availability

The Inside Airbnb dataset reports listing availability over four booking horizons: 30, 60, 90, and 365 days. To provide a consistent measure of short-term availability across all cities, the 90-day availability window is used throughout this analysis.

One limitation is that the dataset records whether a listing is available but does not distinguish nights that have already been booked from nights intentionally blocked by the host. Consequently, lower availability should not be interpreted exclusively as higher booking demand.



Availability varies more substantially across cities than many of the other metrics considered throughout this report. Austin and Chicago exhibit similar distributions, while Los Angeles contains a larger concentration of listings with high availability. Portland falls between these extremes, whereas New York displays the greatest variability.

The borough-level analysis demonstrates that New York is not a homogeneous rental market. Manhattan and Brooklyn exhibit substantially broader availability distributions than the remaining boroughs, helping to explain the unusually large spread observed in the city-wide violin plot.

Availability remains relatively consistent across host categories. Although modest differences exist, median availability differs by only a few days in most cities. Portland provides the clearest exception, where individual hosts exhibit lower median availability than small or large hosts.

Room type likewise exhibits relatively small differences in median availability. Hotel rooms display the greatest variation between cities, with Chicago exhibiting noticeably lower availability than the remaining markets. However, hotel rooms account for less than 1% of all listings, limiting the broader importance of this observation.

Overall, the availability analysis suggests that geographic market conditions are more closely associated with observed listing availability than either host category or room type.

4 Conclusion

This analysis examined Airbnb listings across Austin, Chicago, Los Angeles, New York, and Portland to determine how pricing, host composition, room type, and availability vary across short-term rental markets. Although several broad patterns appeared across the five cities, the results consistently demonstrate that Airbnb markets are strongly shaped by local geographic and market conditions.

Price variation was particularly dependent on location. Most cities contained spatial clusters with relatively similar median nightly prices alongside one or more clusters that departed substantially from the surrounding market. These differences were often associated with geographically distinct areas, including Santa Catalina Island in Los Angeles and more affluent communities surrounding Portland. New York provided the clearest example of the limitations of city-wide summaries: borough-level analysis revealed substantial differences in listing composition and price that were obscured by the broader clustering results.

Host and room composition also revealed several recurring patterns. Individual hosts represented the majority of hosts in every city, while small and large hosts controlled a disproportionately large share of listings relative to their population. Entire homes and apartments dominated most markets, although private rooms were more prevalent in specific locations and among small hosts. This relationship provides one possible explanation for the lower median prices frequently observed among small-host listings.

The regression analysis found that property-related characteristics were more consistent predictors of nightly price than host characteristics. Number of bathrooms produced the largest positive coefficients across all five cities, although it likely also serves as a proxy for unobserved property size. Accommodation capacity was statistically significant but had a smaller association with price, while review score and superhost status varied across markets. These results reinforce the broader finding that no single pricing relationship applies uniformly across every city.

Availability showed a similar pattern. Differences between host and room categories were generally modest, whereas availability distributions varied substantially across cities and, in New York, across boroughs. This suggests that geographic market conditions are more closely associated with observed availability than either host category or room type. However, because the dataset does not distinguish booked nights from dates intentionally blocked by hosts, availability cannot be interpreted as a direct measure of demand.

Several limitations should be considered when interpreting these results. The analysis represents a single period of data and therefore does not capture seasonal or long-term changes in short-term rental markets. The dataset also lacks potentially important property characteristics such as square footage, and several geographic and regulatory explanations remain observational rather than causal. Future work could incorporate multiple time periods, additional cities, neighborhood-level socioeconomic data, and alternative spatial or predictive models.

Overall, the project demonstrates that broad cross-city patterns are useful for identifying common market structures, but local geography remains essential to understanding how those patterns appear in practice. Across pricing, host composition, room type, regression results, and availability, the most consistent conclusion is not that all Airbnb markets behave alike, but that similar market forces produce different outcomes depending on the city and location in which they operate.